Chapter 7 — CTEs, Views and Recursion: Naming Queries and Walking Trees

A report needing "order totals, then customer totals, then customers above average" becomes three unreadable levels of nested subqueries. This chapter covers SQL's tools for **naming** a piece of a query so big queries are built from small, readable steps:

- a **common table expression (CTE)** names a query for the length of *one* statement;
- a **recursive CTE** calls itself, which lets plain SQL walk hierarchies (org charts, category trees, bills of materials — parts lists where each part can itself be made of parts) and graphs (networks of connected items: flight routes, friend networks) and produce sequences;
- a **view** names a query *permanently* in the database, so everyone can reuse it like a table;
- a **materialized view** also stores the query's *result*, trading freshness for speed.

Plus `generate_series`, PostgreSQL's way to produce rows of numbers or dates for "fill in the missing days" reports.

Sections: 7.1 CTE basics · 7.2 Data-modifying CTEs · 7.3 Recursive CTE mechanics · 7.4 Hierarchies · 7.5 Graphs and cycles · 7.6 `generate_series` · 7.7 Views · 7.8 Materialized views · 7.9 Choosing.

Setup. The `company` database (chapter 1, 1.11) plus a `routes` table (a small flight network, 7.5); `code/00_schema.sql` builds everything. Subqueries and derived tables, which CTEs replace: chapter 6.

7.1 Common table expressions (WITH)

What it is

A **common table expression** is a named, temporary result set defined at the start of a statement with `WITH name AS (query)` and used in the rest of that statement like a table — named scrap paper ("let *x* = total per customer") that disappears when the statement ends.

```
WITH cte_name [(col1, col2, ...)] AS (
    SELECT ...                -- any query
)
[, another_cte AS (SELECT ... FROM cte_name ...)]   -- later CTEs may use earlier ones
SELECT ... FROM cte_name ...;  -- the main statement
```

The problem it solves

Nested derived tables (subqueries in `FROM`) read *inside-out* — the first step is buried deepest. CTEs write the logic *top-down* with a name per step, and let one intermediate result be used **several times** without copying it.

How it works

1. Each `name AS (query)` defines one CTE; separate several with commas, write `WITH` once.
2. A CTE can reference CTEs defined **before** it (and, with `RECURSIVE`, itself — 7.3).
3. The final `SELECT`/`INSERT`/`UPDATE`/`DELETE` uses them by name.
4. Nothing is stored in the catalog (the database's internal list of objects); the CTE lives for this one statement.

One CTE — employees paid above their department average:

```sql
-- code/01_cte_basics.sql
WITH dept_avg AS (
    SELECT dept_id, round(avg(salary), 0) AS avg_salary
    FROM employees
    WHERE dept_id IS NOT NULL
    GROUP BY dept_id
)
SELECT e.first_name, e.dept_id, e.salary, d.avg_salary
FROM employees e
JOIN dept_avg d ON d.dept_id = e.dept_id
WHERE e.salary > d.avg_salary
ORDER BY e.dept_id, e.salary DESC;
```

Output:

```
 first_name | dept_id |  salary   | avg_salary
------------+---------+-----------+------------
 Vikram     |       1 | 185000.00 |     130750
 Neha       |       1 | 142000.00 |     130750
 Daniel     |       2 | 150000.00 |      89500
 Lucas      |       3 | 120000.00 |      90500
 Kiran      |       4 | 135000.00 |     105500
(5 rows)
```

`dept_avg` behaves like a table with one row per staffed department.

Several CTEs as a pipeline — order value → customer lifetime value → overall average:

```sql
WITH order_totals AS (          -- step 1: value of each non-cancelled order
    SELECT o.order_id, o.customer_id,
           sum(oi.quantity * oi.unit_price * (1 - oi.discount)) AS order_value
    FROM orders o
    JOIN order_items oi USING (order_id)
    WHERE o.status <> 'cancelled'
    GROUP BY o.order_id, o.customer_id
),
customer_totals AS (            -- step 2: lifetime value per customer
    SELECT customer_id, count(*) AS orders, round(sum(order_value), 2) AS lifetime_value
    FROM order_totals
    GROUP BY customer_id
),
overall AS (                    -- step 3: one number to compare against
    SELECT round(avg(lifetime_value), 2) AS avg_ltv FROM customer_totals
)
SELECT c.first_name, c.last_name, ct.orders, ct.lifetime_value, o.avg_ltv
FROM customer_totals ct
JOIN customers c USING (customer_id)
CROSS JOIN overall o
WHERE ct.lifetime_value > o.avg_ltv
ORDER BY ct.lifetime_value DESC;
```

Output:

```
 first_name | last_name | orders | lifetime_value | avg_ltv
------------+-----------+--------+----------------+---------
 Anita      | Shah      |      3 |        3454.08 | 1057.66
 Ben        | Carter    |      2 |        1858.36 | 1057.66
(2 rows)
```

`overall` has one row, so `CROSS JOIN` attaches the average to every row. Nested subqueries would be three levels deep.

Using a CTE twice — `monthly` is read by the main query *and* a scalar subquery:

```sql
WITH monthly AS (
    SELECT date_trunc('month', placed_at)::date AS month, count(*) AS orders
    FROM orders
    GROUP BY 1
)
SELECT month, orders,
       orders = (SELECT max(orders) FROM monthly) AS is_best_month
FROM monthly
ORDER BY month;
```

Output:

```
   month    | orders | is_best_month
------------+--------+---------------
 2025-01-01 |      3 | t
 2025-02-01 |      2 | f
 ...
 2025-04-01 |      3 | t
 ...
(6 rows)
```

Naming the columns in the header. `WITH dept_size(dept, headcount) AS (SELECT d.name, count(e.emp_id) …)` renames the output columns — handy for unnamed expressions like `count(e.emp_id)` (`code/01_cte_basics.sql`).

How PostgreSQL executes a CTE: inlined or materialized

- **Inlined** — the optimizer (the component choosing how to run a query) pastes the CTE into the main query like a subquery and optimizes everything together (e.g. pushing a `WHERE` down so an index — a sorted lookup structure — can be used).
- **Materialized** — the CTE runs once into a temporary work area that the main query scans; outside filters can't reach inside.

Since **PostgreSQL 12**, a non-recursive, side-effect-free (read-only) CTE referenced **once** is inlined automatically; one referenced two or more times is materialized. Override with `AS MATERIALIZED` / `AS NOT MATERIALIZED`. (Before 12 every CTE was materialized — the "optimization fence" old blog posts warn about.)

```sql
EXPLAIN (COSTS OFF)
WITH big AS (SELECT * FROM employees)
SELECT first_name FROM big WHERE emp_id = 3;

EXPLAIN (COSTS OFF)
WITH big AS MATERIALIZED (SELECT * FROM employees)
SELECT first_name FROM big WHERE emp_id = 3;
```

Output:

```
                  QUERY PLAN
-----------------------------------------------
 Index Scan using employees_pkey on employees
   Index Cond: (emp_id = 3)
(2 rows)

          QUERY PLAN
------------------------------
 CTE Scan on big
   Filter: (emp_id = 3)
   CTE big
     ->  Seq Scan on employees
(4 rows)
```

The first plan never mentions the CTE — the filter reached the primary-key index; the second copies the whole table, then filters. (`EXPLAIN` prints the chosen plan; `Seq Scan` = read the whole table; chapter 12.)

`MATERIALIZED` suits an expensive CTE referenced several times, or something to compute exactly once (a slow function call). `NOT MATERIALIZED` suits a cheap multiply-referenced CTE that each use filters differently.

Dialect notes

Feature	PostgreSQL	MySQL	SQLite	SQL Server
`WITH` (non-recursive)	yes	8.0+	3.8.3+	2005+
`WITH RECURSIVE` keyword	required for recursion	required	required	*not used* — plain `WITH` can recurse
`MATERIALIZED` hint	12+	no	3.35+	no
CTE with `INSERT/UPDATE/DELETE` inside	yes (7.2)	no	no	no (but a CTE can be the *target* of an `UPDATE` / `DELETE`)

In SQL Server the statement *before* a `WITH` must end with a semicolon — hence `;WITH` in T-SQL code (not run).

Trade-offs

	Derived table (subquery in `FROM`)	CTE
Readability	inside-out	top-down, named steps
Reuse in the same statement	copy-paste	reference by name
Recursion	no	yes
Performance (PostgreSQL 12+)	same when inlined	same when inlined; can force materialization
Scope	the one place it's written	the whole statement

When not to use it

- The query is readable without it — one level of subquery is fine.
- Many statements need the same logic: a **view** (7.7).
- The intermediate result is needed across statements in a session (one connection's lifetime), perhaps indexed: a **temporary table** (visible only to your session, dropped when it ends; 7.9).

Common mistakes

- **`WITH` twice** (`WITH a AS (...) WITH b AS (...)`), a missing comma between CTEs, or an extra one after the last.
- **Referencing a CTE defined later** — a CTE sees only those above it (or itself, if recursive).
- **Assuming a CTE "caches" work** — a single-use CTE is inlined (12+); use `MATERIALIZED` for one evaluation.

Real-world examples

Analytics teams chain CTEs (`raw → cleaned → joined → aggregated → final`); dbt (a tool for building data-transformation pipelines from SQL files) encourages this style.

In the interview

30-second answer: "A CTE is a named subquery declared with `WITH` that exists for one statement. It makes complex queries readable as a sequence of steps, lets you reference the same intermediate result several times, and — with `WITH RECURSIVE` — lets a query refer to itself to walk hierarchies. In PostgreSQL 12 and later a CTE referenced once is inlined, so it costs the same as a subquery; you can force materialization with `AS MATERIALIZED`."

- *Is a CTE faster than a subquery?* → Usually identical (inlined). Slower if materialized and a selective outer filter can't be pushed in; faster if an expensive CTE is materialized once and reused.
- *CTE vs temporary table?* → A CTE lasts one statement and isn't indexed; a temp table lasts the session (or transaction), can be indexed and analyzed (statistics gathered for the optimizer), and serves many statements.
- *Can you update through a CTE?* → In PostgreSQL the CTE itself can contain `UPDATE/DELETE/INSERT ... RETURNING` (next section).

Self-check

1. *In PostgreSQL 14, is a CTE referenced once materialized?* → No, inlined — unless `AS MATERIALIZED`, recursive, or with side effects.

7.2 Data-modifying CTEs (PostgreSQL)

What it is

In PostgreSQL a CTE may contain `INSERT`, `UPDATE` or `DELETE` with `RETURNING`, whose rows feed the rest of the statement. Several changes ("delete these rows and copy them to an archive") become **one statement**, automatically atomic — all or nothing.

How it works

1. `DELETE/UPDATE/INSERT ... RETURNING` inside a CTE produces the affected rows; later CTEs or the main statement read them by name.
2. Every data-modifying CTE runs **to completion exactly once**, even if nothing reads its output.
3. All parts see the **same snapshot** (a frozen picture of the data) from *before* the statement — not each other's changes, except through `RETURNING`.

Example — archive cancelled orders and their items:

```sql
-- code/02_data_modifying_cte.sql
CREATE TABLE orders_archive (LIKE orders INCLUDING ALL);
CREATE TABLE order_items_archive (LIKE order_items INCLUDING ALL);

WITH gone_items AS (
    DELETE FROM order_items
    WHERE order_id IN (SELECT order_id FROM orders WHERE status = 'cancelled')
    RETURNING *
),
saved_items AS (
    INSERT INTO order_items_archive SELECT * FROM gone_items RETURNING order_id
),
gone_orders AS (
    DELETE FROM orders
    WHERE status = 'cancelled'
    RETURNING *
)
INSERT INTO orders_archive
SELECT * FROM gone_orders
RETURNING order_id, status;
```

Output:

```
 order_id |  status
----------+-----------
      105 | cancelled
      111 | cancelled
(2 rows)
```

```
SELECT (SELECT count(*) FROM orders)              AS live_orders,
       (SELECT count(*) FROM orders_archive)      AS archived_orders,
       (SELECT count(*) FROM order_items_archive) AS archived_items;
```

Output:

```
 live_orders | archived_orders | archived_items
-------------+-----------------+---------------
          12 |               2 |              2
(1 row)
```

`CREATE TABLE ... (LIKE t INCLUDING ALL)` copies the column definitions, defaults and constraints of `t` (see chapter 9).

The snapshot rule, shown:

```
WITH bump AS (
    UPDATE products SET unit_price = unit_price + 1 WHERE product_id = 2
    RETURNING unit_price
)
SELECT (SELECT unit_price FROM products WHERE product_id = 2) AS seen_by_select,
       (SELECT unit_price FROM bump)                          AS new_value;
```

Output:

```
 seen_by_select | new_value
----------------+-----------
          24.99 |     25.99
(1 row)
```

The outer `SELECT` reads the old price; only `RETURNING` shows the new one. Afterwards the table holds 25.99.

Common mistakes

- **Modifying the same row twice in one statement** (two CTEs) — unpredictable result.
- **Expecting a later CTE to "see" an earlier change via the table** — read the earlier CTE's `RETURNING`.
- **Porting to MySQL/SQL Server** — unsupported; use a transaction (statements committed or undone together) of separate statements (chapter 13).

7.3 Recursive CTEs: the mechanics

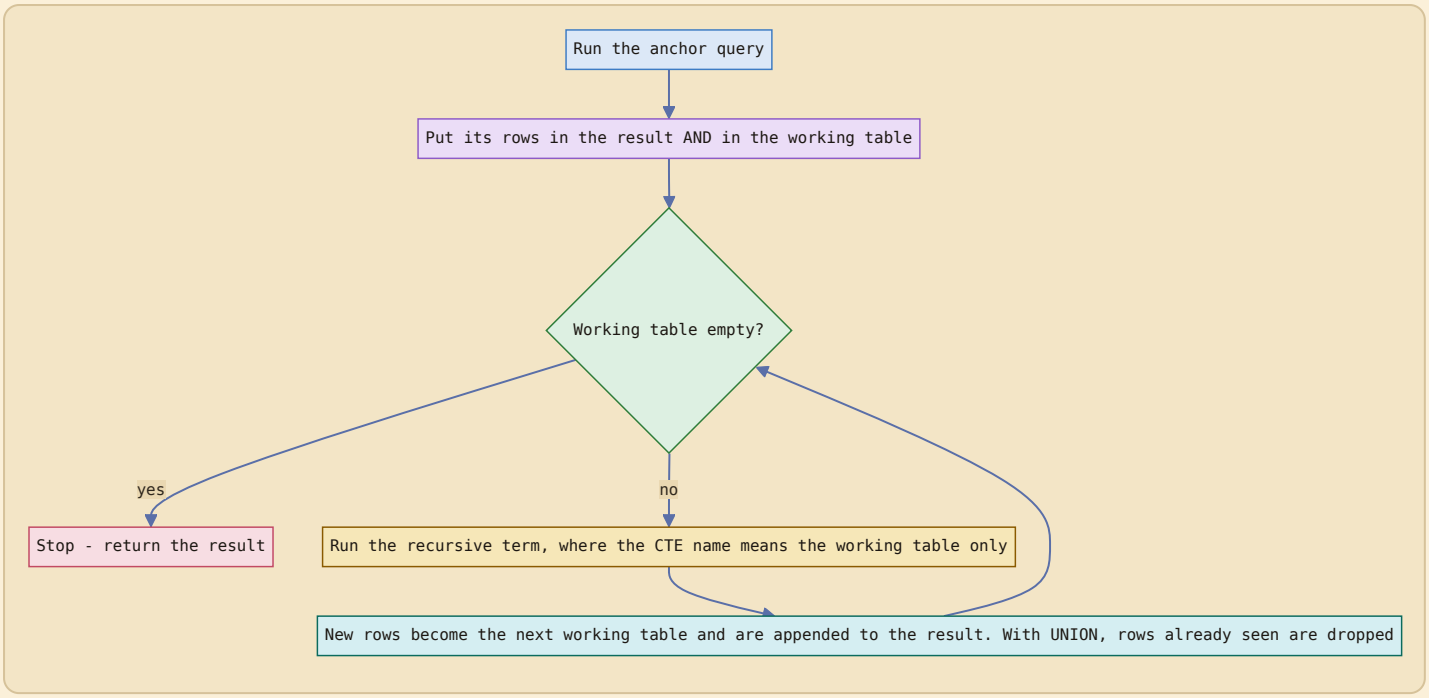
What it is

A **recursive CTE** refers to itself: `WITH RECURSIVE name AS (anchor UNION ALL recursive_term)`. It is standard SQL's *repetition*: "start with these rows, keep deriving new rows from the last batch until none appear" — like listing a grandparent's descendants generation by generation until one has no children.

The problem it solves

Questions with no fixed depth — "all reports under this manager at every level", "every category inside Electronics", "every city reachable from Mumbai". A join handles one level (chapter 5 self-join); a recursive CTE handles any depth.

How it works



1. **Anchor** (before `UNION [ALL]`) runs once, producing the starting rows.
2. **Recursive term** (after it) references the CTE name, which on each pass means *only the previous pass's rows* (the "working table"); its output is appended to the result and feeds the next pass.
3. **Termination:** stop when a pass produces no rows — your `WHERE` or join must make that happen.
4. `UNION ALL` keeps every row; `UNION` discards rows already produced, which also stops loops whose state repeats.

PostgreSQL rules: the recursive term references the CTE only **once**, not inside an aggregate, a subquery, or the nullable side of an outer join. Anchor and recursive term need the same column count and compatible types — cast in the anchor (`0::bigint`, `name::text`) if the recursive term produces a wider type.

Counting 1 to 5:

```sql
-- code/03_recursive_series.sql
WITH RECURSIVE counter(n) AS (
    SELECT 1                -- anchor: the starting row(s)
    UNION ALL
    SELECT n + 1            -- recursive term: runs on the rows produced last time
    FROM counter
    WHERE n < 5             -- stop condition
)
SELECT n FROM counter;
```

Output:

```
 n
---
 1
 2
 3
 4
 5
(5 rows)
```

Trace: anchor → {1}. Pass 1 reads {1} → {2}. Pass 2 reads {2} → {3} … pass 4 reads {4} → {5}. Pass 5 reads {5}; `5 < 5` is false → no rows → stop.

Carrying state in several columns — the next Fibonacci pair (each number the sum of the two before) and a running factorial ($n! = 1 \times 2 \times \ldots \times n$):

```
WITH RECURSIVE fib(i, a, b, fact) AS (
    SELECT 1, 0::bigint, 1::bigint, 1::numeric
    UNION ALL
    SELECT i + 1, b, a + b, fact * (i + 1)
    FROM fib
    WHERE i < 10
)
SELECT i, a AS fibonacci, fact AS factorial FROM fib;
```

Output:

```
 i  | fibonacci | factorial
----+-----------+-----------
  1 |         0 |         1
  2 |         1 |         2
  3 |         1 |         6
  4 |         2 |        24
 ...
 10 |        34 |   3628800
(10 rows)
```

Walking dates works the same way: anchor `SELECT DATE '2025-01-06'`, recursive term `SELECT (week_start + 7)::date FROM weeks WHERE week_start < DATE '2025-02-03'` → five Mondays (`code/03_recursive_series.sql`). PostgreSQL's `generate_series` (7.6) is simpler; the recursive form is for MySQL, which lacks it.

A runaway query and two safety nets. Without a stop condition recursion never ends. While experimenting, an outer `LIMIT` stops it (PostgreSQL evaluates recursion lazily, only as far as the outer query asks):

```
WITH RECURSIVE forever(n) AS (
    SELECT 1 UNION ALL SELECT n + 1 FROM forever      -- no stop condition!
)
SELECT n FROM forever LIMIT 3;
```

Output:

```
 n
---
 1
 2
 3
(3 rows)
```

And `UNION` stops a computation whose state repeats: doubling modulo 7 (remainder after dividing by 7) goes 1 → 2 → 4 → 1; the second 1 is discarded and the working table empties:

```
WITH RECURSIVE mod_cycle(n) AS (
    SELECT 1
    UNION
    SELECT (n * 2) % 7 FROM mod_cycle
)
SELECT n FROM mod_cycle;
```

Output:

```
 n
---
 1
 2
 4
(3 rows)
```

`UNION` only helps when the *whole row* repeats; a growing column (depth, path, running total) never repeats — use explicit cycle detection (7.5).

Dialect notes

	PostgreSQL	MySQL 8	SQLite	SQL Server
Syntax	`WITH RECURSIVE`	`WITH RECURSIVE`	`WITH RECURSIVE`	`WITH` (no keyword)
Built-in depth limit	none	`cte_max_recursion_depth` = 1000 by default	none	100 by default; `OPTION (MAXRECURSION n)`, 0 = unlimited
`UNION` (distinct) in recursion	yes	yes	yes	no, `UNION ALL` only
`SEARCH` / `CYCLE` clauses	14+	no	no	no

Oracle also supports recursive `WITH` (since 11g R2) and its older proprietary `CONNECT BY` syntax.

Common mistakes

- **No stop condition** → runs until memory or disk runs out (MySQL/SQL Server: depth-limit error).
- **Anchor/recursive type mismatch.** The anchor fixes column types: a `varchar(40)` anchor column extended by `path || ' > ' || name` (`text`) fails with *"column has type character varying(40) in non-recursive term but type text overall"*; `SELECT 1` makes an `integer` that a growing factorial overflows. Cast in the anchor (`first_name::text`, `1::numeric`).
- **Thinking the recursive term sees the whole result** — only the previous pass.
- **Relying on output order** — build a sort key (depth, path, or `SEARCH`, 7.5) and `ORDER BY` it.

Self-check

1. *When does recursion stop?* → When a pass produces no new rows.
2. *What does SQL Server need instead of `WITH RECURSIVE`?* → Just `WITH`; recursion is limited to 100 levels unless `OPTION (MAXRECURSION n)`.

7.4 Hierarchies: trees stored as parent pointers

What it is

The usual relational tree is the **adjacency list**: each row stores its parent's key (`employees.manager_id` → another `emp_id`; `NULL` marks the root), queried with recursive CTEs.

How it works — three classic queries

Question	Anchor	Recursive step
Top-down: the whole tree (or a subtree)	the root(s): `manager_id IS NULL` (or `emp_id = X`)	children: `e.manager_id = parent.emp_id`
Bottom-up: the chain of ancestors	the starting node	parent: `m.emp_id = child.manager_id`
Subtree aggregates	every (boss, direct report) pair	extend each pair one level down, keeping the boss

Top-down with depth and path (the path doubles as a sort key keeping each team under its manager):

```sql
-- code/04_hierarchy.sql
WITH RECURSIVE org AS (
    SELECT emp_id, first_name, manager_id, 1 AS depth,
           first_name::text AS path
    FROM employees
    WHERE manager_id IS NULL                        -- anchor: the root
    UNION ALL
    SELECT e.emp_id, e.first_name, e.manager_id, o.depth + 1,
           o.path || ' > ' || e.first_name
    FROM employees e
    JOIN org o ON e.manager_id = o.emp_id           -- children of rows found so far
)
SELECT repeat('    ', depth - 1) || first_name AS org_chart, depth, path
FROM org
ORDER BY path;
```

Output:

```
   org_chart    | depth |         path
----------------+-------+----------------------
 Asha           |     1 | Asha
     Daniel     |     2 | Asha > Daniel
         Arjun  |     3 | Asha > Daniel > Arjun
         Priya  |     3 | Asha > Daniel > Priya
         Sara   |     3 | Asha > Daniel > Sara
     Kiran      |     2 | Asha > Kiran
         Omar   |     3 | Asha > Kiran > Omar
 ...
     Vikram     |     2 | Asha > Vikram
         Li     |     3 | Asha > Vikram > Li
 ...
         Rohan  |     3 | Asha > Vikram > Rohan
(15 rows)
```

Pass 1 found Asha's four reports (depth 2), pass 2 theirs (depth 3), pass 3 nobody — stop. A text path sorts correctly only because names lack `>`; a robust key is an id array (`ARRAY[emp_id]`, then `path || e.emp_id`), compared element by element. For one subtree, anchor on `WHERE emp_id = 2`.

Bottom-up: who is above the intern?

```sql
WITH RECURSIVE chain AS (
    SELECT emp_id, first_name, job_title, manager_id, 0 AS steps_up
    FROM employees WHERE emp_id = 15
    UNION ALL
    SELECT m.emp_id, m.first_name, m.job_title, m.manager_id, c.steps_up + 1
    FROM employees m
    JOIN chain c ON m.emp_id = c.manager_id
)
SELECT steps_up, first_name, job_title FROM chain ORDER BY steps_up;
```

Output:

```
 steps_up | first_name |      job_title
----------+------------+---------------------
        0 | Rohan      | Intern
        1 | Vikram     | Engineering Manager
        2 | Asha       | CEO
(3 rows)
```

Only the join direction differs: `m.emp_id = c.manager_id` ("find my parent") vs `e.manager_id = o.emp_id` ("find my children"). Uses: breadcrumbs (`Home > Electronics > Laptops`), approval chains, permission inheritance.

Subtree totals — carry the *boss* unchanged while walking down, one row per (boss, anyone below):

```sql
WITH RECURSIVE reports AS (
    SELECT manager_id AS boss, emp_id, salary        -- direct reports
    FROM employees WHERE manager_id IS NOT NULL
    UNION ALL
    SELECT r.boss, e.emp_id, e.salary                -- their reports, and so on
    FROM reports r
    JOIN employees e ON e.manager_id = r.emp_id
)
SELECT b.first_name AS manager, count(*) AS team_size, sum(r.salary) AS team_salary
FROM reports r JOIN employees b ON b.emp_id = r.boss
GROUP BY b.first_name
ORDER BY team_size DESC, manager;
```

Output:

```
 manager | team_size | team_salary
---------+-----------+-------------
 Asha    |        14 |  1367000.00
 Vikram  |         5 |   432000.00
 Daniel  |         3 |   208000.00
 Kiran   |         1 |    76000.00
 Lucas   |         1 |    61000.00
(5 rows)
```

Asha has all 14 others below her; Vikram has Neha, Rahul, Meera, Li, Rohan. The aggregate runs *outside* — aggregates aren't allowed in the recursive term.

Other ways to store trees (for comparison)

Model	How	Reads	Writes	Notes
Adjacency list (this chapter)	`parent_id` column	recursive CTE	trivial (change one pointer)	the default choice
Materialized path	store `'1/2/15'` (or PostgreSQL `ltree`, an extension type for dotted label paths)	`LIKE '1/2/%'`, indexable	moving a subtree rewrites all descendants' paths	great for breadcrumbs
Nested sets	store `lft`/`rgt` numbers	subtree = range query	inserts renumber many rows	read-heavy, rarely-changing trees
Closure table	separate table of every (ancestor, descendant, depth) pair	simple joins	many rows per insert	*SQL Antipatterns* recommends it for heavy tree querying

Common mistakes

- **Anchoring on `manager_id = X`** lists X's reports but *not X*; use `emp_id = X` to include X.
- **Cyclic bad data** (A manages B, B manages A): a walk from a real root never reaches it, but one starting inside loops forever. Protect with a path check (7.5) or a `CHECK`/trigger (function run automatically on row changes) preventing cycles.
- **Sorting by depth only** loses the "team under manager" layout; sort by path.

In the interview

30-second answer: "Store the tree as an adjacency list — each row points to its parent. To read a subtree, anchor on the root and recursively join children whose parent is in the previous level, carrying depth and a path for ordering. For ancestors, reverse the join. For totals per node, carry the top node through the recursion and aggregate outside."

- *How would you stop infinite loops if the data could contain a cycle?* → Carry an array of visited ids and add `WHERE NOT e.emp_id = ANY(path)`, or use PostgreSQL 14's `CYCLE` clause.

- *Huge tree queried constantly?* → Index `manager_id`; consider a closure table or materialized path (`ltree` with a GiST index — a PostgreSQL index type for not-simply-sortable data like paths or shapes).

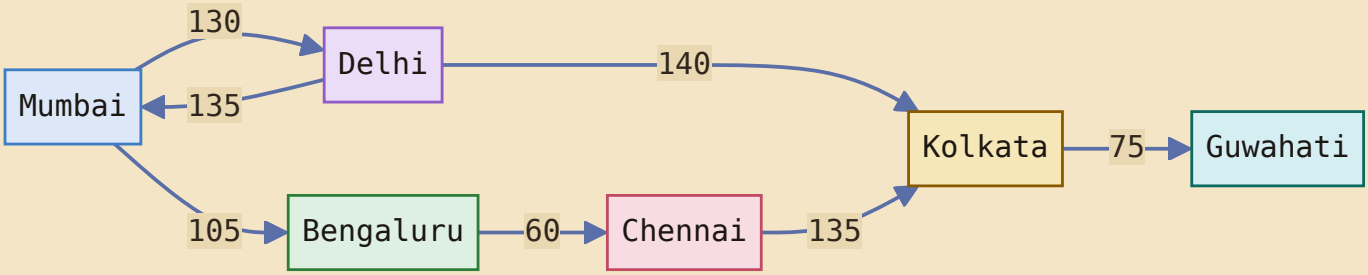
Self-check

1. *Which join walks down, which up?* → Down: `child.manager_id = found.emp_id`; up: `parent.emp_id = found.manager_id`.

7.5 Graphs and cycles

What it is

A **graph** is a tree whose nodes can have several parents and whose paths can loop back (a **cycle**): flight routes, roads, "follows", package dependencies. `routes` stores one-way flights; Mumbai → Delhi → Mumbai is a cycle.



The problem it solves

A naive walk loops forever (Mumbai → Delhi → Mumbai …), and each row differs (total time grows), so `UNION` won't stop it. Remember where you have been:

How it works (manual cycle detection)

carry an **array of visited nodes** (`path`) and follow an edge only if its destination isn't in it (`WHERE r.to_city <> ALL (t.path)`). Every row is then a *simple path* (no repeated city); there are finitely many, so recursion ends.

```sql
-- code/05_graph_cycles.sql
WITH RECURSIVE trips AS (
    SELECT to_city AS city, ARRAY['Mumbai', to_city] AS path, minutes AS total_min
    FROM routes WHERE from_city = 'Mumbai'
    UNION ALL
    SELECT r.to_city, t.path || r.to_city, t.total_min + r.minutes
    FROM trips t
    JOIN routes r ON r.from_city = t.city
    WHERE r.to_city <> ALL (t.path)              -- never revisit a city
)
SELECT city, array_to_string(path, ' -> ') AS route, total_min
FROM trips
ORDER BY city, total_min;
```

Output:

```
   city    |                          route                          | total_min
-----------+---------------------------------------------------------+-----------
 Bengaluru | Mumbai -> Bengaluru                                     |       105
 Chennai   | Mumbai -> Bengaluru -> Chennai                          |       165
 Delhi     | Mumbai -> Delhi                                         |       130
 Guwahati  | Mumbai -> Delhi -> Kolkata -> Guwahati                  |       345
 Guwahati  | Mumbai -> Bengaluru -> Chennai -> Kolkata -> Guwahati   |       375
 Kolkata   | Mumbai -> Delhi -> Kolkata                              |       270
 Kolkata   | Mumbai -> Bengaluru -> Chennai -> Kolkata               |       300
(7 rows)
```

`ARRAY[...]` builds an array (chapter 14), `||` appends, `x <> ALL (array)` is true when `x` differs from every element (chapter 6). Delhi → Mumbai is never followed: Mumbai is in every path.

Shortest (fastest) route to each city — keep the best row per city with `DISTINCT ON` (chapter 2):

```sql
-- same trips CTE as above, then:
SELECT DISTINCT ON (city) city, total_min, array_to_string(path, ' -> ') AS route
FROM trips
WHERE city <> 'Mumbai'
ORDER BY city, total_min;
```

Output:

```
   city    | total_min |                route
-----------+-----------+--------------------------------------
 Bengaluru |       105 | Mumbai -> Bengaluru
 Chennai   |       165 | Mumbai -> Bengaluru -> Chennai
 Delhi     |       130 | Mumbai -> Delhi
 Guwahati  |       345 | Mumbai -> Delhi -> Kolkata -> Guwahati
 Kolkata   |       270 | Mumbai -> Delhi -> Kolkata
(5 rows)
```

This enumerates *every* simple path — exponential work (can double per extra hop) on big, dense graphs. Fine for small graphs and interviews; real route planners use shortest-path algorithms (Dijkstra, A* — never enumerate every path) in code or extensions such as pgRouting, or graph databases (store nodes and edges directly; chapter 16).

The CYCLE and SEARCH clauses (PostgreSQL 14+)

SQL:1999 helper clauses, implemented in PostgreSQL 14 (this book's version):

- `CYCLE col SET is_cycle USING path_col` — tracks visited `col` values, flags a row whose value was already on its path with `is_cycle = true`, and stops recursing from it.
- `SEARCH DEPTH FIRST BY col SET ord` / `SEARCH BREADTH FIRST BY col SET ord` — adds `ord` to `ORDER BY` for depth-first (one branch fully down before the next) or breadth-first (level by level) order.

```sql
WITH RECURSIVE trips(city, total_min) AS (
    SELECT to_city, minutes FROM routes WHERE from_city = 'Mumbai'
    UNION ALL
    SELECT r.to_city, t.total_min + r.minutes
    FROM trips t JOIN routes r ON r.from_city = t.city
) CYCLE city SET is_cycle USING visited
SELECT city, total_min, is_cycle FROM trips
WHERE city IN ('Mumbai', 'Delhi')
ORDER BY total_min;
```

Output:

```
  city  | total_min | is_cycle
--------+-----------+----------
 Delhi  |       130 | f
 Mumbai |       265 | f
 Delhi  |       395 | t
(3 rows)
```

Mumbai (265) is *not* flagged: the anchor starts from *destinations*, so Mumbai wasn't yet on the tracked path; the hop back to Delhi repeats Delhi and is flagged. To count the start city as visited, include it in the anchor (as `ARRAY['Mumbai', to_city]` did).

```
WITH RECURSIVE org(emp_id, first_name, manager_id) AS (
    SELECT emp_id, first_name, manager_id FROM employees WHERE manager_id IS NULL
    UNION ALL
    SELECT e.emp_id, e.first_name, e.manager_id
    FROM employees e JOIN org o ON e.manager_id = o.emp_id
) SEARCH BREADTH FIRST BY emp_id SET ord
SELECT first_name, ord FROM org ORDER BY ord LIMIT 6;
```

Output:

```
 first_name |  ord
------------+--------
 Asha       | (0,1)
 Vikram     | (1,2)
 Daniel     | (1,6)
 Lucas      | (1,10)
 Kiran      | (1,12)
 Neha       | (2,3)
(6 rows)
```

Breadth-first `ord` is (level, id); `DEPTH FIRST` produces an array path like our manual `path`.

Self-check

1. *Why doesn't `UNION` stop the Mumbai–Delhi loop?* → Each lap has a different `total_min`/path, so rows never repeat exactly.

7.6 generate_series: making rows (PostgreSQL)

What it is

`generate_series(start, stop [, step])` is a PostgreSQL **set-returning function**: in `FROM` it returns one row per value from `start` to `stop`, for integers, numerics and timestamps/dates (step as an `interval`).

```
-- code/06_generate_series.sql
SELECT * FROM generate_series(1, 5) AS n;
SELECT * FROM generate_series(10, 0, -5) AS n;
SELECT d::date AS day FROM generate_series(DATE '2025-03-01', DATE '2025-03-04', INTERVAL '1 day') AS d;
```

Output (the three results, one after another):

```
 n
---
 1
 2
 ...
 5
(5 rows)

 n
----
 10
  5
  0
(3 rows)

    day
------------
 2025-03-01
 2025-03-02
 2025-03-03
 2025-03-04
(4 rows)
```

With `date` arguments the `timestamp with time zone` version is used, so cast back to `date`.

The problem it solves: missing rows in reports

`GROUP BY` only makes groups for values that **exist**: a month with no orders vanishes and a chart silently skips it. Generate the full calendar, `LEFT JOIN` the data, count:

```sql
WITH months AS (
    SELECT m::date AS month
    FROM generate_series(DATE '2025-01-01', DATE '2025-07-01', INTERVAL '1 month') AS m
)
SELECT m.month, count(o.order_id) AS orders
FROM months m
LEFT JOIN orders o ON date_trunc('month', o.placed_at) = m.month
GROUP BY m.month
ORDER BY m.month;
```

Output:

```
   month    | orders
------------+--------
 2025-01-01 |      3
 2025-02-01 |      2
 ...
 2025-06-01 |      2
 2025-07-01 |      0
(7 rows)
```

July shows `0` — via `count(o.order_id)`; `count(*)` would count July's `NULL`-padded row as 1 (chapter 4).

Test data in one statement:

```sql
CREATE TABLE fake_events AS
SELECT g AS id, DATE '2025-01-01' + (g % 90) AS day, (g * 37) % 100 AS score
FROM generate_series(1, 1000) AS g;
SELECT count(*), min(day), max(day), round(avg(score), 1) AS avg_score FROM fake_events;
```

Output:

```
 count |    min     |    max     | avg_score
-------+------------+------------+-----------
  1000 | 2025-01-01 | 2025-03-31 |      49.5
(1 row)
```

Arithmetic instead of `random()` keeps data deterministic (identical every run) for reproducible performance tests (chapter 12 scales this to millions of rows).

Dialect notes

	Numbers / dates series
PostgreSQL	`generate_series(...)`
SQL Server 2022+	`GENERATE_SERIES(start, stop [, step])` — numbers only (SQL Server — not run)
SQLite	`generate_series` table-valued function from the `series` extension (built into the `sqlite3` shell)
MySQL 8	none — use a recursive CTE (7.3) or a permanent "numbers"/"calendar" table

Data warehouses (databases for analysing large historical data) often keep a permanent **calendar table** (one row per date with flags like `is_weekend`, `fiscal_quarter`) for this.

7.7 Views

What it is

A **view** is a named query stored in the database, queried like a table (`SELECT * FROM staff_directory`) but storing **no data**: each use substitutes its query into yours and runs it against the current **base** tables — like a saved email search, always up to date.

Why it matters

Benefit	Example
Simplicity — hide a complex join once	analysts query `staff_directory` instead of re-writing the join
Security — expose only some columns/rows	grant access to a view without `salary`, not to `employees`
Stable interface — change tables, keep the view	rename/split a table and redefine the view so old queries still work
Consistency — one definition of a metric	"active customer" is defined once, not in 20 reports

How it works

`CREATE VIEW name AS query;` stores the parsed query in the catalog. When a query mentions the view, PostgreSQL's rewriter (a pre-planning stage that expands views) substitutes it — like an inlined CTE — and the optimizer plans the whole. So a view is always current and costs exactly its query.

```sql
-- code/07_views.sql
CREATE VIEW staff_directory AS
SELECT e.emp_id, e.first_name || ' ' || e.last_name AS full_name,
       e.job_title, d.name AS department, e.email
FROM employees e
LEFT JOIN departments d USING (dept_id);

SELECT full_name, job_title, department FROM staff_directory
WHERE department = 'Sales' ORDER BY full_name;
```

Output:

```
   full_name   |   job_title   | department
---------------+---------------+------------
 Arjun Singh   | Sales Rep     | Sales
 Daniel Brooks | Sales Manager | Sales
 Priya Nair    | Sales Rep     | Sales
 Sara Khan     | Sales Rep     | Sales
(4 rows)
```

Always current: after `UPDATE employees SET job_title = 'Senior Sales Rep' WHERE emp_id = 7`, the view shows Priya Nair as Senior Sales Rep immediately.

Changing and dropping views

- `CREATE OR REPLACE VIEW` swaps the definition in place (keeping permissions), but in PostgreSQL only if it returns the **same columns, names, types and order**; you may only **add** columns at the end.
- To remove or reorder columns: `DROP VIEW` + `CREATE VIEW` (re-granting privileges given with `GRANT`).
- Dependencies are tracked: a table (or view) used by a view can't be dropped without `CASCADE`, which drops the dependent views too.

```sql
-- adding hire_date at the end: allowed
CREATE OR REPLACE VIEW staff_directory AS
SELECT e.emp_id, e.first_name || ' ' || e.last_name AS full_name,
       e.job_title, d.name AS department, e.email, e.hire_date
FROM employees e
LEFT JOIN departments d USING (dept_id);

-- removing columns: refused
CREATE OR REPLACE VIEW staff_directory AS
SELECT e.emp_id, e.job_title FROM employees e;

-- dropping a table the view depends on: refused
DROP TABLE departments;
```

Output:

```
ERROR:  cannot drop columns from view
ERROR:  cannot drop table departments because other objects depend on it
DETAIL:  constraint employees_dept_id_fkey on table employees depends on table departments
constraint projects_dept_id_fkey on table projects depends on table departments
view staff_directory depends on table departments
HINT:  Use DROP ... CASCADE to drop the dependent objects too.
```

(`CASCADE` would also drop the two foreign-key constraints — rarely what you want.)

Updatable views

Writes through a view work if each view row maps to exactly one base row. Since PostgreSQL 9.3 a view is **automatically updatable** when it reads exactly **one** table or updatable view and has no `GROUP BY`, `HAVING`, `DISTINCT`, `LIMIT`/`OFFSET`, window functions, aggregates, `UNION`/`INTERSECT`/`EXCEPT`, or top-level `WITH`. Computed columns are read-only. Otherwise write an `INSTEAD OF` trigger (a function run automatically in place of the write; chapter 14).

```sql
CREATE VIEW furniture AS
SELECT product_id, name, unit_price, category
FROM products
WHERE category = 'Furniture';

UPDATE furniture SET unit_price = 329.99 WHERE name = 'Ergonomic Chair';
INSERT INTO furniture VALUES (10, 'Bookshelf', 120.00, 'Furniture');
-- Nothing stops a row that falls OUTSIDE the view's WHERE...
INSERT INTO furniture VALUES (11, 'Webcam', 59.00, 'Electronics');
SELECT product_id, name, category FROM products WHERE product_id >= 10;
SELECT count(*) AS visible_in_view FROM furniture WHERE product_id >= 10;
```

Output:

```
 product_id |   name    |  category
------------+-----------+-------------
         10 | Bookshelf | Furniture
         11 | Webcam    | Electronics
(2 rows)

 visible_in_view
-----------------
               1
(1 row)
```

The Webcam went into `products` through `furniture` but is invisible there — a "disappearing row". (`discontinued`, not in the view, got its default `false`.) `WITH CHECK OPTION` rejects writes whose row would not satisfy the view's `WHERE`:

```sql
CREATE OR REPLACE VIEW furniture AS
SELECT product_id, name, unit_price, category
FROM products
WHERE category = 'Furniture'
WITH CHECK OPTION;

INSERT INTO furniture VALUES (12, 'Monitor Arm', 79.00, 'Accessories');
```

Output:

```
ERROR:  new row violates check option for view "furniture"
DETAIL:  Failing row contains (12, Monitor Arm, Accessories, 79.00, f).
```

(`LOCAL` checks only this view's condition; `CASCADED`, the default, also those of underlying views.) A view with a join or `GROUP BY` is read-only:

```sql
CREATE VIEW dept_payroll AS
SELECT d.name AS department, count(e.emp_id) AS headcount, sum(e.salary) AS payroll
FROM departments d LEFT JOIN employees e USING (dept_id)
GROUP BY d.name;
SELECT * FROM dept_payroll ORDER BY payroll DESC NULLS LAST;
UPDATE dept_payroll SET headcount = 0 WHERE department = 'Sales';
```

Output:

```
 department  | headcount |  payroll
-------------+-----------+-----------
 Engineering |         4 | 523000.00
 Sales       |         4 | 358000.00
 ...
 Research    |         0 |    (null)
(6 rows)

ERROR:  cannot update view "dept_payroll"
DETAIL:  Views containing GROUP BY are not automatically updatable.
HINT:  To enable updating the view, provide an INSTEAD OF UPDATE trigger or an unconditional ON UPDATE DO INSTEAD rule.
```

The catalog knows which views are updatable:

```sql
SELECT table_name, is_updatable
FROM information_schema.views
WHERE table_schema = 'public'
ORDER BY table_name;
```

Output:

```
   table_name    | is_updatable
-----------------+--------------
 dept_payroll    | NO
 furniture       | YES
 staff_directory | NO
(3 rows)
```

Views and security

A PostgreSQL view reads base tables with the **view owner's** privileges, so `GRANT SELECT ON staff_directory TO analyst` exposes names and titles but never salaries, with no grant on `employees`.

- `WITH (security_barrier)` stops a crafted user function in the caller's `WHERE` from peeking at rows the view's `WHERE` hides (otherwise the optimizer may run the cheap function first).
- PostgreSQL 15 added `WITH (security_invoker = true)` to use the *caller's* privileges (not run here on 14).

Roles, `GRANT` and row-level security (per-user row limits): chapter 14.

Dialect notes

	PostgreSQL	MySQL	SQLite	SQL Server
`CREATE OR REPLACE VIEW`	yes (add columns only)	yes (any change)	no — drop & create	`CREATE OR ALTER VIEW` (2016 SP1+)
Auto-updatable simple views	yes	yes	no — read-only, use `INSTEAD OF` triggers	yes
`WITH CHECK OPTION`	yes	yes	no	yes
Materialized views	yes (7.8)	no	no	"indexed views" (kept up to date automatically)

When not to use it

- **Deep view stacks** — each looks simple, the combined query is huge and hard to tune.
- **As a performance tool** — a view is exactly as fast as its query; use a materialized view or indexes.
- **`SELECT *` in a definition** — `*` is expanded at *creation*, so later table columns do **not** appear. List columns.

Common mistakes

- Expecting a view to store data (that's a materialized view).
- Relying on `ORDER BY` inside a view — only the **outer** query's `ORDER BY` guarantees order.

Real-world examples

BI (business-intelligence) tools (Metabase, Tableau, Power BI) point at views, not raw tables; after splitting `users` into `accounts` and `profiles`, a `users` view keeps old code working; PostgreSQL's `information_schema` is views over the system catalogs.

In the interview

30-second answer: "A view is a stored query that acts like a virtual table. It stores no data, so it's always current and costs the same as running its query. Views simplify complex joins, provide a stable interface when tables change, and enforce security by exposing only certain columns or rows. Simple single-table views are updatable; `WITH CHECK OPTION` stops writes that would fall outside the view."

- *Does a view improve performance?* → Not by itself; it is expanded into the query. A materialized view (or indexed view in SQL Server) can.

- *Can you insert into a view?* → Yes if it's a simple single-table view (auto-updatable) or has `INSTEAD OF` triggers.
- *What happens to a view if you add a column to its table?* → Nothing; the view keeps the column list captured when it was created.

Self-check

1. *Why did the Webcam vanish from `furniture`, and what prevents it?* → It doesn't satisfy `category = 'Furniture'`; `WITH CHECK OPTION`.
2. *Why is `staff_directory` not updatable?* → It joins two tables.

7.8 Materialized views

What it is

A **materialized view** stores its query's *result* on disk like a table and recomputes only on `REFRESH MATERIALIZED VIEW` — trading **freshness** for **speed** (like a printed report: instant to read, stale until reprinted).

How it works

1. `CREATE MATERIALIZED VIEW name AS query;` runs the query and stores the rows (`WITH NO DATA` creates it empty, unqueryable until refreshed). Reads can use indexes you create on it.
2. `REFRESH MATERIALIZED VIEW name;` re-runs and replaces the contents under a strong lock (others wait): **reads are blocked**.
3. `REFRESH MATERIALIZED VIEW CONCURRENTLY name;` builds the new result separately and applies only differences, not blocking readers; needs a **unique index** (plain columns, no `WHERE`) and is slower overall.

```sql
-- code/08_materialized_views.sql
CREATE MATERIALIZED VIEW product_sales AS
SELECT p.product_id, p.name,
       coalesce(sum(oi.quantity), 0) AS units_sold,
       coalesce(round(sum(oi.quantity * oi.unit_price * (1 - oi.discount)), 2), 0) AS revenue
FROM products p
LEFT JOIN order_items oi USING (product_id)          -- every order line
GROUP BY p.product_id, p.name;

SELECT name, units_sold, revenue FROM product_sales ORDER BY revenue DESC LIMIT 3;
```

Output:

```
      name       | units_sold | revenue
-----------------+------------+---------
 Laptop Pro 14   |          5 | 6380.05
 Standing Desk   |          2 |  978.00
 Ergonomic Chair |          3 |  962.47
(3 rows)
```

Stale until refreshed:

```sql
INSERT INTO orders VALUES (115, 9, NULL, '2025-06-20 09:00', 'pending', NULL);
INSERT INTO order_items VALUES (115, 5, 3, 499.00, 0.00);

SELECT name, units_sold, revenue FROM product_sales WHERE name = 'Standing Desk';
REFRESH MATERIALIZED VIEW product_sales;
SELECT name, units_sold, revenue FROM product_sales WHERE name = 'Standing Desk';
```

Output:

```
     name      | units_sold | revenue
---------------+------------+---------
 Standing Desk |          2 |  978.00
(1 row)

     name      | units_sold | revenue
---------------+------------+---------
 Standing Desk |          5 | 2475.00
(1 row)
```

Concurrent refresh needs a unique index:

```
REFRESH MATERIALIZED VIEW CONCURRENTLY product_sales;          -- fails
CREATE UNIQUE INDEX product_sales_pk ON product_sales (product_id);
REFRESH MATERIALIZED VIEW CONCURRENTLY product_sales;          -- works
CREATE INDEX product_sales_revenue ON product_sales (revenue DESC);
SELECT matviewname, ispopulated FROM pg_matviews;
```

Output:

```
ERROR:  cannot refresh materialized view "public.product_sales" concurrently
HINT:  Create a unique index with no WHERE clause on one or more columns of the materialized view.
  matviewname  | ispopulated
---------------+-------------
 product_sales | t
(1 row)
```

Trade-offs

	View	Materialized view	Summary table maintained by triggers/jobs
Stores data	no	yes	yes
Freshness	always current	as of last refresh	as current as your code keeps it
Read speed	cost of the query	fast (indexable)	fast
Write overhead	none	none at write time; refresh cost later	on every write
Effort	trivial	trivial + schedule refreshes	most code, most bugs

PostgreSQL has **no automatic or incremental refresh** — each refresh recomputes everything. Schedule it (cron, the Unix timetable scheduler; the `pg_cron` extension; a job scheduler) or run it after batch loads. Oracle has incremental "fast refresh"; SQL Server's indexed views update on every write (many restrictions); MySQL has none.

When to use / not use

- **Use** for dashboards/reports that may be minutes or hours old, expensive aggregations read far more than the data changes, precomputed lookup data.
- **Don't** when readers need up-to-the-second numbers, or refreshing costs more than it saves.

Common mistakes

- No scheduled refresh — the dashboard shows last month's numbers.
- Plain `REFRESH` on a busy materialized view, blocking every reader.

In the interview

30-second answer: "A view stores a query; a materialized view stores the query's result. Materialized views make expensive aggregations fast to read and can be indexed, but they're stale

> until refreshed. In PostgreSQL refresh is manual and full; `CONCURRENTLY` avoids blocking readers but needs a unique index."

- *Keeping it fresh?* → Scheduled refresh (e.g. every 5 minutes with pg_cron) or after each ETL batch (Extract-Transform-Load job copying and reshaping data in); for real-time, maintain a summary table incrementally.
- *vs a cache like Redis?* → The materialized view is in the database, SQL-queryable, consistent as of a point in time; an external cache (hot data outside the database; Redis is an in-memory key-value store) is faster per key but needs invalidation (updating copies when data changes; System Design Fundamentals, chapter 15).

Self-check

1. *What does `CONCURRENTLY` need?* → A unique index on the materialized view with no `WHERE` clause.

7.9 Choosing the right tool

Need	Use	Lifetime	Stores rows?	Indexable?
One-off intermediate result inside one query, used once	subquery / derived table (ch. 6)	one statement	no	no
Readable multi-step query, reuse within the statement, recursion, move rows atomically	CTE	one statement	only if materialized, temporarily	no
Reusable definition for many queries and users, security, stable interface	view	permanent	no	no (index the base tables)
Fast reads of expensive results that may be slightly stale	materialized view	permanent	yes	yes
Intermediate data reused by several statements in a session, needs indexes/statistics	`CREATE TEMP TABLE` (ch. 9)	session or transaction	yes	yes

---

Chapter 7 TL;DR

- **CTE** (`WITH name AS (...)`) names a step for one statement; chain with commas; reuse by name. PostgreSQL 12+ inlines single-use CTEs; `MATERIALIZED` / `NOT MATERIALIZED` override.
- **Data-modifying CTEs** (PostgreSQL) run `INSERT/UPDATE/DELETE ... RETURNING` inside `WITH` — atomic multi-step changes; all parts see the pre-statement snapshot.
- **Recursive CTE** = anchor `UNION [ALL]` recursive term; the recursive term sees only the previous pass's rows; stops when a pass returns nothing. Cast types in the anchor, always have a stop condition, `ORDER BY` a path/depth.
- **Trees**: adjacency list + recursion — join children to go down, join the parent to go up, carry the top node and aggregate outside for subtree totals.
- **Graphs**: carry a visited array and refuse revisits, or use `CYCLE` (PostgreSQL 14+); `SEARCH BREADTH/DEPTH FIRST` gives an ordering column.
- **`generate_series`** makes number/date rows: full calendar + `LEFT JOIN` + `count(col)` fills gaps with zeros. MySQL uses a recursive CTE instead.
- **View** = stored query, no data, always current; for simplicity, security, stable interfaces. Simple single-table views are updatable; `WITH CHECK OPTION` blocks out-of-view writes; `CREATE OR REPLACE` can only append columns in PostgreSQL.

- **Materialized view** = stored result; fast, indexable, stale until `REFRESH` (`CONCURRENTLY` needs a unique index). No auto-refresh in PostgreSQL.

Next: chapter 8 — window functions, which compute running totals, rankings and comparisons with neighbouring rows without collapsing rows the way `GROUP BY` does.